

GREEN HOUSE MONITORING AND CONTROL SYSTEM USING IOT

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ABSTRACT

Greenhouse monitoring and Control System is the approach in which the rural areas farmers will be benefitted by this Greenhouse environment. This project focuses on the Generic architecture and it can also be applied for many other automation-based application. Direct supervision of human will be replaced. By referring the different papers, we have proposed the system based on the limitations which is present in the monitoring and control system. Greenhouse is where the plants are grown with the controlled manner. Nowadays, Greenhouse construction is in greater demand due to the lack of land availability and urbanization. Growing crops will be reserved mainly by this proposed Greenhouse project. From the location wireless, Greenhouse using IoT can be used to monitor and control many Greenhouse buildings.

Keywords: Atmospheric Sensors, Iot, Arduino UNO, Website Or Cloud, DC Motor, Relay.

I. INTRODUCTION

A Greenhouse, which is a exceptionally outlined homestead structure. This gives a more controllable environment to a better crop security, transplantation, harvest generation and product seeding. As a part of this present-day period, for developing yields, following to more space of area which has vigorously utilized for commercial ventures and housing the area space is accessible. The cost-effective farming such as new blossoms, organic products and vegetables generation is the utilization of Greenhouse development in many tropical nations. The conformity of ideal atmosphere development conditions which is to attain to the high return at great quality, low natural burden and low cost is the effectiveness of plant creation inside Greenhouse which depends fundamentally. Parameters like light, humidity, temperature, soil moisture must be controlled ideally where the given certain criteria through water creation, warming, ventilation and lighting are used to attain certain objectives. By persistent checking and controlling of these ecological parameters which gives a significant data that is related to the individual impacts of the elements differently towards acquiring the most extreme creation of harvest. The present remarkable difficulties of Greenhouse is to control. Temperature changes quickly, In Greenhouse as a result, fluctuations relying upon sun powered radiation level, moisture levels and outside temperatures. Poor natural product set and quality frequently bought by the high stickiness and poor light intensity. Expanding the efficiency of labourers by empowering them for the more important assignments, electrical expenses and heating fuel, empowering producers and directors to settle on better administration choices and to invest more on the energy dealing with procedures can be decreased by practicing exact control over the system.

In this thesis, we have proposed a system which can read the parameters data from the environment which is identified within the Greenhouse and will be controlled manually or by Android app which we have developed. In the cloud data will be stored for future use. If any climate values fluctuate the precautionary measures will be taken by the user. The relationship between reference estimations and sensor flags, breaking down the developments, the natural variables which are uncovered and advancement of yields are explored by strictly observing climate changing conditions.

II. METHODOLOGY

The main objective of this project is to control the system automatically. The temperature sensor, light sensor, co2 sensor and humidity sensor will be controlled. By implementing advance technology in the industry will increase the demand for agriculture products which are concerned by the consumer's needs.

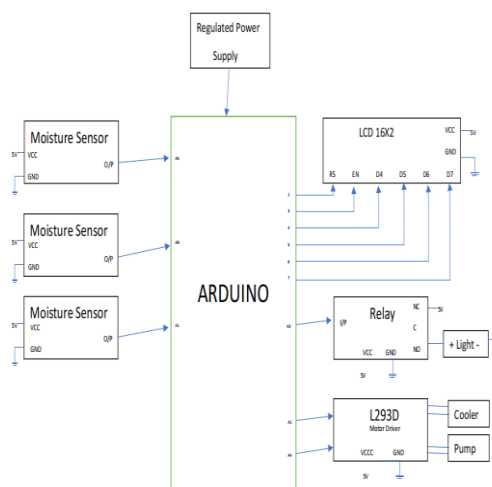


Figure 1: Block diagram of Greenhouse Monitoring and Control System Using IOT

III. MODELING AND ANALYSIS



Figure 1: Light sensor

A light sensor is a photoelectric device that converts light energy (photons) detected to electrical energy (electrons).



Figure 2: Temperature sensor

Temperature sensor is the device which converts temperature into an electrical signal and which can be a Thermocouple which provides electrical signals that is readable from the temperature measurement



Figure 3: Moisture sensor

Soil moisture sensors refer to sensors which estimate volumetric water content.

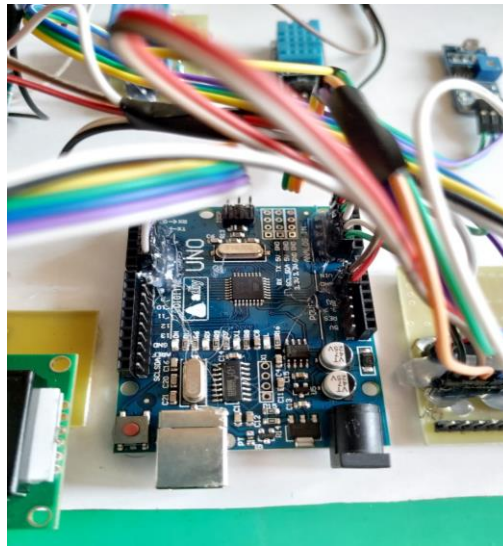


Figure 4: Arduino UNO Board

Arduino UNO is a categorized as a microcontroller which uses ATmega328 as the microcontroller in it and used for an electronic project and which are preferred by the beginners.

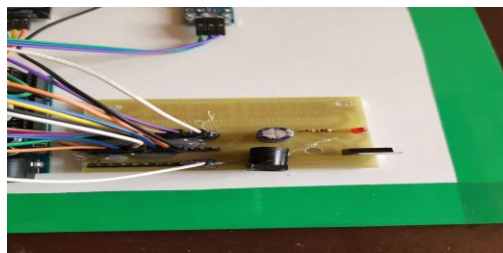


Figure 5: Power supply

5V Power supply is required for the entire control system.



Figure 6: Display console

Moisture and temperature measured values will be displayed on this console.

IV. RESULTS AND DISCUSSION

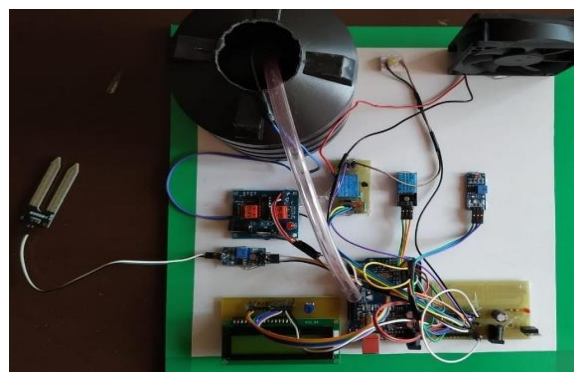


Figure 1: Greenhouse control system

This control unit collects all the details regarding the plant growth. Consisting of moisture, temperature and light sensor with a mini water tank attached for supplying sufficient amount of water to plants. The whole controlling system is having a power supply of 5v.

LED is present in the controlling unit for providing enough light for the plants and a mini fan for controlling the temperature. Display console shows the measured values of moisture and temperature. We have created a cloud for storing all the details about the growth of plants. And have Developed an android application for monitoring the greenhouse and controlling the environment inside the greenhouse.

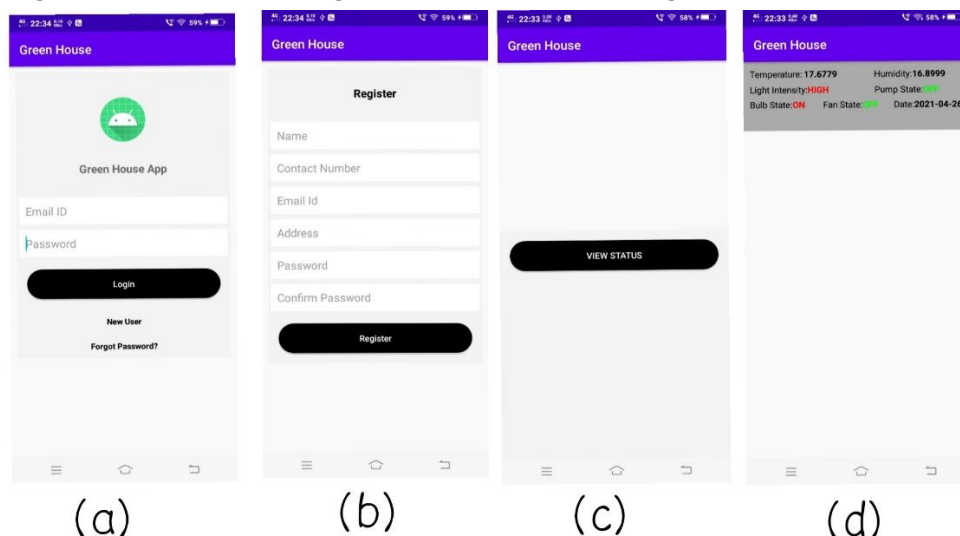


Figure 2 : Mobile Application Screen Flow

(a). This is the home page of mobile application for greenhouse monitoring system, where user can log in to system. (b). This is the registration page for new users. (c). This will be the home page for users to view status of the greenhouse. (d). In this page we can view the parameter values inside the green house.

V. CONCLUSION

This Greenhouse project is designed intelligently for the betterment of the crop growth. The system gives the high efficiency plant growth, low cost and effective working. The user can control it manually and also using the Android app which we have developed. Greenhouse is more effective than the plants grown normally. The Android app is used to check the status inside the Greenhouse and can be handled it easily. The software and hardware working is so smoothly done and can be used easily. The system design give the user to understand the working and very ease to use it. The yield is so effective and the data can be stored in cloud for the future use.

VI. REFERENCES

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